EPSR-test-ex22 nwts: second-order differences (three samples)

Program(s): nwts | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises the three-sample path of `nwts` (`nsamples 3`): from two isotopically distinct samples and their mixture the scattering can be separated into XX, XY and YY partial weights, where Y is the substituted species (here hydrogen) and X all other atoms - the standard second-order difference analysis of H/D substitution experiments that EPSR can refine against [1,2].

Code exercised

Fortran (original, fortran_src/):

- `nwts.f`: `nwts` (third sample from `xfrac`, `<run>tot1/2/3.wts`, `<run>xx/xy/yy.wts`, intra/intermolecular XX/XY/YY branches for exchanging and non-exchanging substituted atoms)
- `process_symbols.f`: `check_symbol` (`nwts` branch: `nsamples` clamped to 1-3, `normtot`)
- `ato_manipulation_rev4.f`: `scatt_len`; `read_and_save_files.f`: `read_input_file`, `read_ato`, `save_files_others`; `ato_read_and_save.f`: `minitc`, `moloutc`

Rust (port, rust/epsr-core/src/):

- `tools/nwts.rs`: `run` (second-order block writing the `xx`/`xy`/`yy` files), `check_symbol_extra`, `read_isotopes`, `scatt_len`
- `symbols.rs`: `read_input_file`, `save_inp_file`; `ato.rs`: `read_ato`, `minitc`, `moloutc`

Physical system

The EPSR-test mixture of 200 DMSO and 400 water molecules (3200 atoms; atom types S, C, O, methyl H, water O1 and H1). Three hypothetical H/D experiments: `SecondDiffA` - sample 1 fully deuterated (H and exchangeable H1 = 2H), sample 2 protiated, sample 3 = 0.4 × sample 1 + 0.6 × sample 2, weights per molecule, not normalised; `SecondDiffB` - per atom, `xfrac 0.65`, <b²>-normalised, with 13C on C, methyl H natural and non-exchanging H1 30 % 1H / 70 % 2H in sample 2; `SecondDiffC` - `xfrac 0.5`, <b>²-normalised. Scattering lengths from Sears [3], Faber-Ziman pair weights [4], in the spirit of the DMSO/water substitution study of Soper and Luzar [5].

Inputs

`SecondDiff{A,B,C}.NWTS.dat` made from `EPSR-test/DMSODWaterD.NWTS.dat` by changing keyword values; `SecondDiffC` asks for `nsamples 4`. `200DMSO400Water.ato` from EPSR-test.

How the accuracy was verified

`commands.txt` (3 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

One pass of three `nwts` calls; 19 files compared: for each input `tot1`, `tot2`, `tot3`, `xx`, `xy` and `yy .wts` files, plus the rewritten `SecondDiffC.NWTS.dat`, in which the Fortran has replaced `nsamples 4` by 3 (the other two input files are rewritten unchanged). `SecondDiffB` reaches the intramolecular YY/XY/XX branches for substituted atoms that do not exchange.

References

[1] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[2] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[3] V. F. Sears, "Neutron scattering lengths and cross sections", Neutron News 3(3), 26-37 (1992).
[4] T. E. Faber and J. M. Ziman, "A theory of the electrical properties of liquid metals III. The resistivity of binary alloys", Phil. Mag. 11, 153-173 (1965).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.